

Nested Innovation Geometry and Projected Dynamics in Hierarchical Shell-Manifold Theory

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October 2026

Abstract

We develop the transition geometry of Hierarchical Shell-Manifold Theory (HSMT) without replacing its graded-manifold formulation. Isometric embeddings $J_N : \mathcal{H}_N \rightarrow \mathcal{H}_{N+1}$ define an innovation sector $\mathcal{K}_{N+1} = \mathcal{H}_{N+1} \ominus J_N \mathcal{H}_N$. This separates inherited structure from new structure and supplies a covariant hierarchy difference, an innovation-weighted quadratic form, and a Feshbach description of projected dynamics.

Localized radial states give an exact Gaussian overlap benchmark. Fixed Gaussian localization, canonical Gram orthogonalization, ordinary harmonic Dirichlet-to-Neumann response, running Gaussian widths, and separable spatial Clifford multiplicity do not independently generate the fractional boundary scaling considered here.

For a semi-infinite hierarchy, the unilateral shift has a finite lower-boundary defect. In a minimal Pauli–Jacobi realization the Feshbach-weighted structural polarization is exactly $\mathfrak{p}_N = -1/(2N - 1)$, so the structural hierarchy anomaly is $\alpha_H^{\min} = -1/2$. This is an N^{-1} hierarchy imbalance without non-Hermiticity. It does not select the candidate fractional coefficient.

If a general anomaly α_H exists and the hierarchy coordinate satisfies $N(\rho) \sim \rho^D$, a positive-frame transport convention assigns the exponent $\nu_H = (D/2)|\alpha_H|$. Under further assumptions—weighted half-line branches $\beta_{\pm} = 1 \pm 2\nu_H$, an effectively one-dimensional innovation spectrum, and a leading p^2 factor in the self-energy—one obtains $J(M) \sim M^{-4\nu_H}$ and $E \sim p^{2/(1+4\nu_H)}$. The values $\nu_H = 1/12$, $c_{\text{IS}} = 35/144$, and $E \sim p^{3/2}$ are one conditional chain. Their microscopic selection reduces to the unresolved calculation of α_H from the full HSMT operator.

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1 Introduction

Hierarchical Shell-Manifold Theory is organized around a graded geometric arena

$$\mathcal{M} = \bigsqcup_N \mathcal{M}_N \quad (1)$$

together with hierarchy and spatial operators whose projected dynamics define an effective observable sector. The architecture used here is

$$\mathcal{M} \longrightarrow D_\rho, D_{\text{sp}} \longrightarrow \text{states and localized packets} \longrightarrow \langle N|M \rangle \longrightarrow \mathcal{D} \longrightarrow P_N \longrightarrow H_{\text{eff}}. \quad (2)$$

This paper does not replace that architecture. It makes explicit several structures already implicit in it, in the sense of the September 2026 framework [1].

Three questions are central. If the labels N distinguish successive grades, what map connects $\mathcal{M}_N$ to $\mathcal{M}_{N+1}$? How should structure inherited from a lower grade be separated from structure that is new at a higher grade? How can that innovation sector modify the dynamics seen after projection?

Throughout,

$$N \neq d, \quad \text{shell} \neq \text{dimension}. \quad (3)$$

The index N labels structural organization. Effective dimension is a measured quantity and may depend on the observable used to define it. Geometric, resolution, spectral, and dynamical dimensions are not assumed equal.

The central construction is a nested sequence of isometric embeddings

$$J_N : \mathcal{H}_N \longrightarrow \mathcal{H}_{N+1}, \quad (4)$$

and the innovation sector

$$\mathcal{K}_{N+1} = \mathcal{H}_{N+1} \ominus J_N \mathcal{H}_N. \quad (5)$$

A second result concerns a lowest grade. For a semi-infinite hierarchy the unilateral shift is not unitary. Its lower-boundary defect survives Feshbach weighting and, in a minimal model, produces a normalized structural polarization proportional to N^{-1} . That mechanism fixes the *form* of a possible fractional hierarchy exponent. It does not fix the numerical value previously considered in HSMT calculations.

We therefore define, whenever the limit exists,

$$\alpha_H = \lim_{N \rightarrow \infty} N \frac{\text{Tr}(\Pi_N \Gamma_h A \Pi_N)}{\text{Tr}(\Pi_N A \Pi_N)}, \quad A = V K^{-2} V^\dagger, \quad (6)$$

with Γ_h the hierarchy orientation. No value of α_H is inserted by construction.

2 Nested immanence and hierarchy embeddings

2.1 Containment without identifying grade and dimension

Ordinary nested structures supply a geometric motivation:

$$\text{point} \subset \text{line} \subset \text{surface} \subset \text{volume}.$$

A higher-dimensional object may contain lower-dimensional substructures without those substructures ceasing to exist. HSMT generalizes the containment relation. It does not identify the grade N with an ordinary dimension.

Geometric embeddings

$$\iota_N : \mathcal{M}_N \hookrightarrow \mathcal{M}_{N+1} \quad (7)$$

encode structural inheritance. The disjoint union (1) distinguishes the grades as elements of the hierarchy. The two structures play different roles.

At a common physical location x , the nested structures may be written schematically as

$$\mathcal{E}_1(x) \subset \mathcal{E}_2(x) \subset \mathcal{E}_3(x) \subset \cdots . \quad (8)$$

We call this nested immanence. “Nested” refers to containment. “Immanent” means that the structures need not be separated in ordinary physical space. The coordinates are distinguished:

$$x^\mu = \text{physical location}, \quad \rho = \text{hierarchy coordinate}. \quad (9)$$

A separation $|\rho_N - \rho_M|$ is not assumed to be an ordinary spatial distance.

2.2 Hilbert-space embeddings

At the state-space level we take isometric maps

$$J_N : \mathcal{H}_N \longrightarrow \mathcal{H}_{N+1}, \quad J_N^\dagger J_N = I_N. \quad (10)$$

The new structure at grade $N + 1$ is the orthogonal complement

$$\mathcal{K}_{N+1} = \mathcal{H}_{N+1} \ominus J_N \mathcal{H}_N, \quad (11)$$

so

$$\mathcal{H}_{N+1} = J_N \mathcal{H}_N \oplus \mathcal{K}_{N+1}. \quad (12)$$

Iterating gives a decomposition into innovation sectors,

$$\mathcal{H}_N \simeq \tilde{\mathcal{K}}_1 \oplus \mathcal{K}_2 \oplus \cdots \oplus \mathcal{K}_N, \quad (13)$$

where the tilde denotes the transported lowest sector. This is not a direct sum of the complete lower-grade Hilbert spaces, which would double-count inherited structure.

For $m > n$, the composite embedding is

$$J_{m \leftarrow n} = J_{m-1} J_{m-2} \cdots J_n, \quad (14)$$

and the inherited image of grade n inside grade m is selected by

$$P_n^{(m)} = J_{m \leftarrow n} J_{m \leftarrow n}^\dagger. \quad (15)$$

3 Covariant hierarchy differences

States at neighboring grades belong to different Hilbert spaces, so the naive difference $\psi_{N+1} - \psi_N$ is not intrinsically defined.

Definition 1 (Hierarchy difference) *The covariant discrete hierarchy difference is*

$$\nabla_N \psi = \psi_{N+1} - J_N \psi_N. \quad (16)$$

It vanishes when the state at grade $N + 1$ is purely inherited. If

$$\psi_{N+1} = J_N \tilde{\psi}_N + \eta_{N+1}, \quad \eta_{N+1} \in \mathcal{K}_{N+1},$$

then

$$\nabla_N \psi = J_N (\tilde{\psi}_N - \psi_N) + \eta_{N+1}. \quad (17)$$

The hierarchy difference therefore separates changes within inherited structure from new components.

A natural quadratic form is

$$Q_\rho[\Psi] = \sum_N c_{N+1/2} \|\psi_{N+1} - J_N \psi_N\|^2. \quad (18)$$

The corresponding hierarchy Laplacian is formally $L_\rho = \nabla_\rho^\dagger \nabla_\rho$. The structural chain is

$$\text{nested containment} \longrightarrow J_N \longrightarrow \nabla_N \longrightarrow Q_\rho \longrightarrow L_\rho.$$

In a continuum limit one may write $J_N \simeq I + a_N \mathcal{A}_\rho$, so that

$$\nabla_\rho = \partial_\rho + \mathcal{A}_\rho. \quad (19)$$

A weighted continuum realization has the form

$$L_\rho = -\frac{1}{w(\rho)} \nabla_\rho [w(\rho) \nabla_\rho] + U(\rho). \quad (20)$$

This is a possible geometric origin for the weighted half-line operators used below. It does not by itself select the weight.

4 Localized radial states and Gaussian overlap

4.1 Radial Dirac benchmark

Consider

$$D_\rho = i\sigma^1 \partial_\rho + W(\rho)\sigma^2 + m_0\sigma^3. \quad (21)$$

For $W(\rho) = \omega(\rho - \rho_0)$,

$$D_\rho^2 = -\partial_\rho^2 + \omega^2(\rho - \rho_0)^2 + m_0^2 - \omega\sigma^3, \quad (22)$$

with spectrum

$$M_n^2 = m_0^2 + 2\omega n, \quad n = 0, 1, 2, \dots \quad (23)$$

For $m_0 = 0$ the benchmark contains an exact zero mode.

4.2 Localized packets

The normalized Gaussian ground state centered at ρ_N is

$$g_N(\rho) = \left(\frac{\omega}{\pi}\right)^{1/4} \exp\left[-\frac{\omega}{2}(\rho - \rho_N)^2\right]. \quad (24)$$

The exact overlap is

$$S_{NM} = \langle g_N | g_M \rangle = \exp \left[-\frac{\omega}{4} (\rho_N - \rho_M)^2 \right]. \quad (25)$$

For equally spaced centers $\rho_N = Na_\rho$,

$$S_{NM} = \exp \left[-\frac{\omega a_\rho^2}{4} (N - M)^2 \right]. \quad (26)$$

Comparison with a phenomenological kernel $\exp[-(N - M)^2/(2\sigma_0^2)]$ gives

$$\sigma_0^2 = \frac{2}{\omega a_\rho^2}. \quad (27)$$

This relates Gaussian packet parameters to the phenomenological width. It does not select a numerical value for σ_0 .

4.3 Unequal widths

For running local curvature,

$$S_{NM} = \left(\frac{2\sqrt{\omega_N \omega_M}}{\omega_N + \omega_M} \right)^{1/2} \exp \left[-\frac{\omega_N \omega_M}{2(\omega_N + \omega_M)} (\rho_N - \rho_M)^2 \right]. \quad (28)$$

Running widths modify transition strengths. That fact alone does not generate a preferred orientation of the hierarchy.

5 Nonorthogonal frames and Gram correction

The localized packets are not radial eigenstates and are not mutually orthogonal. Let Φ be the synthesis map and $G = \Phi^\dagger \Phi$, so $G_{NM} = S_{NM}$. The canonical orthonormalized frame is

$$|\tilde{N}\rangle = G^{-1/2} |N\rangle. \quad (29)$$

For a fixed-width equally spaced Gaussian hierarchy, $G_{NM} = e^{-x(N-M)^2/4}$ with $x = \omega a_\rho^2$. Its Toeplitz symbol is

$$g_x(k) = \sum_{m \in \mathbb{Z}} e^{-xm^2/4} e^{ikm} = \vartheta_3\left(\frac{k}{2}, e^{-x/4}\right). \quad (30)$$

After canonical orthogonalization, the derivative symbol is

$$B_{\text{orth}}(k) = -\frac{i\omega a_\rho}{2} \partial_k \ln g_x(k). \quad (31)$$

Poisson resummation gives

$$g_x(k) = \sqrt{\frac{4\pi}{x}} \sum_{n \in \mathbb{Z}} \exp \left[-\frac{(k - 2\pi n)^2}{x} \right]. \quad (32)$$

Near $k = 0$ in the continuum-overlap regime, $B_{\text{orth}}(k) \simeq ik/a_\rho$. Canonical Gram correction therefore reconstructs an ordinary first-order hierarchy derivative. It does not independently generate a fractional interface factor such as $M^{-1/6}$:

$$\text{Gaussian frame orthogonalization} \not\simeq M^{-1/6}. \quad (33)$$

6 Innovation operators and Feshbach projection

Relative to the inherited/innovation decomposition, write

$$D_{N+1} = \begin{pmatrix} J_N D_N J_N^\dagger + \delta D_N & V_N \\ V_N^\dagger & K_{N+1} \end{pmatrix}. \quad (34)$$

The off-diagonal operator V_N measures dynamical communication between inherited and innovation sectors. The exact Feshbach effective operator is

$$D_{N,\text{eff}}(z) = D_N + \delta D_N + V_N(z - K_{N+1})^{-1} V_N^\dagger. \quad (35)$$

At low energy relative to a gapped innovation sector, the leading response is controlled by

$$A_N = V_N K_{N+1}^{-2} V_N^\dagger. \quad (36)$$

For a single innovation level of gap Δ_N and coupling v_N , $A_N = |v_N|^2/\Delta_N^2$. In the weak-mixing regime the state novelty satisfies $\mathcal{N}_N \simeq 2|v_N|^2/\Delta_N^2$. For several innovation states,

$$\mathcal{N}_N \simeq 2 \sum_a \frac{|v_{aN}|^2}{\Delta_{aN}^2}. \quad (37)$$

This motivates the coupling spectral density

$$J_N(M) = \sum_a |v_{aN}|^2 \delta(M - \Delta_{aN}), \quad (38)$$

and the self-energy

$$\Sigma_N(z) = \int_0^\infty \frac{J_N(M)}{z - M} dM. \quad (39)$$

7 Pauli orientation of hierarchy transitions

The radial Dirac structure supplies a conjugate orientation pair through $\sigma_\pm = (\sigma^1 \pm i\sigma^2)/2$. A neighboring transition can be written schematically as a sum of raising and lowering pieces. Hermiticity requires both orientations to appear with conjugate couplings. It does not select a preferred bulk direction.

Running Gaussian widths produce $O(N^{-1})$ corrections to transition strengths when $\omega_N \sim N^s$. Hermitian reversal exchanges the two conjugate Pauli orientations, so those corrections are orientation-even unless additional structure distinguishes the two directions. Self-adjoint running Gaussian geometry therefore does not by itself produce a net hierarchy orientation.

8 Semi-infinite hierarchy and the boundary defect

8.1 Unilateral hierarchy shift

Let $\mathcal{H}_h = \ell^2(\mathbb{N}_0)$ with basis $|0\rangle, |1\rangle, |2\rangle, \dots$. The unilateral shift $S|N\rangle = |N+1\rangle$ has adjoint $S^\dagger|0\rangle = 0$ and $S^\dagger|N\rangle = |N-1\rangle$ for $N \geq 1$. Therefore

$$S^\dagger S = I, \quad S S^\dagger = I - P_0, \quad (40)$$

where $P_0 = |0\rangle\langle 0|$, and

$$S^\dagger S - S S^\dagger = P_0. \quad (41)$$

The trace of the defect is 1. For a bilateral hierarchy $N \in \mathbb{Z}$ the shift is unitary and the defect vanishes. The effect is a property of a hierarchy with a lower endpoint.

8.2 Minimal Pauli–Jacobi transition operator

Consider

$$V = t(S \otimes \sigma_+ + S^\dagger \otimes \sigma_-). \quad (42)$$

This operator is self-adjoint. Take the stationary innovation operator $K = \Delta I$ and set $\lambda^2 = |t|^2/\Delta^2$. Then

$$A = V K^{-2} V^\dagger = \lambda^2 [S S^\dagger \otimes P_+ + S^\dagger S \otimes P_-], \quad (43)$$

where $P_\pm = \frac{1}{2}(I \pm \sigma^3)$. Using (40),

$$A = \lambda^2 [(I - P_0) \otimes P_+ + I \otimes P_-]. \quad (44)$$

Equivalently,

$$A = \lambda^2 \left[I \otimes I - \frac{1}{2} P_0 \otimes I - \frac{1}{2} P_0 \otimes \sigma^3 \right]. \quad (45)$$

The orientation-odd component is localized entirely at the lower hierarchy boundary.

9 Exact structural polarization

Let $\Pi_N = \sum_{n=0}^{N-1} |n\rangle\langle n|$ project onto the first N grades of the semi-infinite operator. Replacing S by a finite-dimensional truncated shift would create an artificial upper boundary whose index cancels the lower-boundary defect. That truncation is not used here.

With $\Gamma_h = \sigma^3$, the accumulated oriented response and the total transition-active content are

$$Q_N = \text{Tr}[\Pi_N \Gamma_h A \Pi_N] = -\lambda^2, \quad (46)$$

$$C_N = \text{Tr}[\Pi_N A \Pi_N] = \lambda^2(2N - 1). \quad (47)$$

The normalized structural polarization is therefore

$$\mathfrak{p}_N = \frac{Q_N}{C_N} = -\frac{1}{2N - 1}. \quad (48)$$

As $N \rightarrow \infty$,

$$\mathfrak{p}_N = -\frac{1}{2N} - \frac{1}{4N^2} + O(N^{-3}). \quad (49)$$

A finite boundary defect divided by linearly accumulating transition-active content produces an N^{-1} hierarchy imbalance. The ratio t/Δ cancels. This is the central benchmark of the paper.

10 The structural hierarchy anomaly

Definition 2 (Structural hierarchy anomaly) *Let $A = V K^{-2} V^\dagger$ be the Feshbach-weighted innovation metric and let Γ_h be the hierarchy orientation operator. Whenever the limit exists,*

define

$$\alpha_H = \lim_{N \rightarrow \infty} N \frac{\text{Tr}[\Pi_N \Gamma_h A \Pi_N]}{\text{Tr}[\Pi_N A \Pi_N]}, \quad A = V K^{-2} V^\dagger. \quad (50)$$

For the minimal unilateral Pauli model,

$$\alpha_H^{\min} = -\frac{1}{2}. \quad (51)$$

The word “anomaly” here means only this asymptotic structural imbalance. It is not a claim of a quantum-field-theoretic anomaly.

Suppose $N(\rho) \sim \rho^D$. Then $\ln N = D \ln \rho + O(1)$. For a small orientation ratio, one convenient positive-frame convention is

$$\delta_N = \frac{1}{4} \ln \left(\frac{1 + \mathfrak{p}_N}{1 - \mathfrak{p}_N} \right) = \frac{\mathfrak{p}_N}{2} + O(\mathfrak{p}_N^3). \quad (52)$$

This factor $1/4$ is part of the transport convention, not a theorem about the Master Operator. If $\mathfrak{p}_N = \alpha_H/N + O(N^{-2})$, then

$$\sum_{n \leq N} \delta_n = \frac{\alpha_H}{2} \ln N + O(1) = \frac{D\alpha_H}{2} \ln \rho + O(1).$$

This motivates the hierarchy-frame exponent

$$\nu_H = \frac{D}{2} |\alpha_H|, \quad (53)$$

and the asymptotic dressing $\mathcal{U}(\rho) \sim \rho^{\nu_H \sigma^3}$. Equation (53) is a definition of the transport exponent associated with this convention.

11 Exact holonomy in the minimal benchmark

For $\mathfrak{p}_N = -1/(2N - 1)$ the same convention gives, for $N > 1$,

$$\delta_N = \frac{1}{4} \ln \frac{N - 1}{N}. \quad (54)$$

The cumulative transport from $N_0 > 1$ to N telescopes:

$$\mathcal{U}_{N,N_0}^{(+)} = \left(\frac{N_0 - 1}{N} \right)^{1/4}. \quad (55)$$

Hence $\mathcal{U}_N^{(\pm)} \sim N^{\mp 1/4}$. With $N \sim \rho^D$,

$$|\nu_H^{\min}| = \frac{D}{4}. \quad (56)$$

The benchmark demonstrates the boundary mechanism. It does not select a small fractional exponent.

12 Weighted innovation geometry

The weighted half-line operator

$$L_\beta = -\rho^{-\beta} \frac{d}{d\rho} \left(\rho^\beta \frac{d}{d\rho} \right) \quad (57)$$

acts on $L^2((0, \infty), \rho^\beta d\rho)$. The unitary flattening map $U_\beta f = \rho^{\beta/2} f$ gives

$$U_\beta L_\beta U_\beta^{-1} = -\partial_\rho^2 - \frac{\beta(2-\beta)}{4\rho^2}. \quad (58)$$

The branches β and $2-\beta$ share the same inverse-square coefficient. If one *defines* the branches by the transport exponent,

$$\beta_\pm = 1 \pm 2\nu_H, \quad (59)$$

then $\beta_\pm(2-\beta_\pm) = 1 - 4\nu_H^2$, and the flat-measure operator is

$$H_{\text{inv}} = -\partial_\rho^2 + \frac{\nu_H^2 - \frac{1}{4}}{\rho^2}. \quad (60)$$

The Bessel index of this operator is $\nu_B = \nu_H$, and

$$c_{\text{IS}} = \frac{1}{4} - \nu_H^2. \quad (61)$$

The weight is not derived from the minimal Pauli model. It is the weighted problem whose Bessel index has been set equal to the transport exponent (53). That identification is an assumption of the conditional chain.

13 Boundary Weyl response

For

$$H_\nu = -\frac{d^2}{d\rho^2} + \frac{\nu^2 - \frac{1}{4}}{\rho^2} \quad (62)$$

and Euclidean parameter $\kappa^2 > 0$, a decaying solution is $\psi_\kappa(\rho) = \sqrt{\rho} K_\nu(\kappa\rho)$. For noninteger ν , the small- ρ expansion contains the two Frobenius powers $\rho^{1/2 \pm \nu}$, with coefficient ratio

$$\frac{B_\nu(\kappa)}{A_\nu(\kappa)} = \frac{\Gamma(-\nu)}{\Gamma(\nu)} \left(\frac{\kappa}{2} \right)^{2\nu}. \quad (63)$$

Up to normalization and boundary-map conventions, the Weyl function therefore scales as $\kappa^{2\nu}$. The inverse interface response then behaves as

$$v(M) \sim M^{-2\nu_H}, \quad (64)$$

once the spectral parameter is identified with the innovation label M . Scale invariance alone does not select ν .

14 Innovation spectral measure

If the innovation sector has spectral density $\rho_{\text{innov}}(M)$ and boundary factor $v(M)$, define

$$J(M) = \rho_{\text{innov}}(M) |v(M)|^2. \quad (65)$$

Effective one-dimensional innovation propagation gives the simplifying assumption $\rho_{\text{innov}}(M) \sim M^0$, hence

$$J(M) \sim M^{-4\nu_{\text{H}}}. \quad (66)$$

If instead $\rho_{\text{innov}}(M) \sim M^{d_{\perp}-1}$, then $J(M) \sim M^{d_{\perp}-1-4\nu_{\text{H}}}$. The one-dimensional case is a structural assumption. It is not an identification of the physical spacetime dimension.

15 Projected effective dynamics

Suppose the spatial coupling enters the projected self-energy with a leading factor p^2 . Then

$$\Sigma(E, p) \sim p^2 \int_0^\infty \frac{J(M)}{E + M} dM. \quad (67)$$

For $J(M) = CM^{-4\nu_{\text{H}}}$ and $0 < 4\nu_{\text{H}} < 1$,

$$\int_0^\infty \frac{M^{-4\nu_{\text{H}}}}{E + M} dM = E^{-4\nu_{\text{H}}} \frac{\pi}{\sin(4\pi\nu_{\text{H}})}. \quad (68)$$

Hence $\Sigma(E, p) \sim p^2 E^{-4\nu_{\text{H}}}$. If this nonanalytic term controls the low-energy pole, $E \sim p^2 E^{-4\nu_{\text{H}}}$, so

$$E \sim p^z, \quad z = \frac{2}{1 + 4\nu_{\text{H}}} = \frac{2}{1 + 2D|\alpha_{\text{H}}|}. \quad (69)$$

Equation (69) is the bridge between the structural anomaly and projected dynamics under the assumptions stated above. It is not a spectral theorem for the Master Operator.

16 Combined radial and spatial operator

Consider the separable combined operator

$$\mathcal{D} = D_\rho \otimes I + \sigma^3 \otimes D_{\text{sp}}. \quad (70)$$

For $m_0 = 0$, $\{\sigma^3, D_\rho\} = 0$. If the spatial operator commutes with radial variables,

$$\mathcal{D}^2 = D_\rho^2 \otimes I + I \otimes D_{\text{sp}}^2. \quad (71)$$

Ordinary Clifford compatibility does not fix ν_{H} . If $m_0 \neq 0$, then $\{D_\rho, \sigma^3\} = 2m_0 I$ and

$$\mathcal{D}^2 = D_\rho^2 + D_{\text{sp}}^2 + 2m_0 D_{\text{sp}}. \quad (72)$$

The massless radial reference point is distinguished by exact separability of the squared operator.

For a factorized hierarchy projector $P = P_{\text{h}} \otimes I_{\text{sp}}$, one finds $V = V_\rho \otimes I_{\text{sp}}$. Spatial degeneracy multiplies numerator and denominator of α_{H} and cancels. Separable spatial Clifford multiplicity therefore does not dilute the structural anomaly. A genuinely spatial contribution requires a

nonseparable projector,

$$[P_N, I \otimes D_{\text{sp}}] \neq 0. \quad (73)$$

Whether such a projector follows from the full HSMT spectral construction is open.

17 The conditional fractional chain

The values previously considered in HSMT calculations form one chain, conditional on the first member.

If $\nu_H = \nu_B = 1/12$, then $\beta_{\pm} = 5/6$ and $7/6$, and

$$c_{\text{IS}} = \frac{1}{4} - \frac{1}{144} = \frac{35}{144}. \quad (74)$$

The inverse boundary response is $v(M) \sim M^{-1/6}$. For a flat one-dimensional innovation density, $J(M) \sim M^{-1/3}$, so $\Sigma(E, p) \sim p^2 E^{-1/3}$ and $E \sim p^{3/2}$. In sequence,

$$\begin{aligned} \nu_B = \frac{1}{12} &\implies \beta_{\pm} = \frac{5}{6}, \frac{7}{6} \implies c_{\text{IS}} = \frac{35}{144}, \\ &\implies v(M) \sim M^{-1/6} \implies J(M) \sim M^{-1/3}, \\ &\implies \Sigma(E, p) \sim p^2 E^{-1/3} \implies E \sim p^{3/2}. \end{aligned} \quad (75)$$

These values are not independent numerical inputs. The first member is not selected by the constructions of this paper.

Using $\nu_H = (D/2)|\alpha_H|$, the condition for this chain is

$$|\alpha_H| = \frac{1}{6D}. \quad (76)$$

The further candidate $D = 2$ would require $|\alpha_H| = 1/12$. Neither $D = 2$ nor $|\alpha_H| = 1/12$ is derived here.

18 The minimal model as a counterexample

The minimal semi-infinite Pauli–Jacobi model gives $|\alpha_H^{\min}| = 1/2$, hence $\nu_H^{\min} = D/4$. At $D = 1$ this is $\nu_H = 1/4$, so $c_{\text{IS}} = 3/16$ and $z = 1$. At $D = 2$,

$$\nu_H^{\min} = \frac{1}{2}, \quad (77)$$

which is the critical inverse-square value, not $1/12$. A lowest grade therefore supplies the $1/N$ form and, in this benchmark, the wrong coefficient:

$$\text{lowest-grade defect} \not\sim \nu_H = \frac{1}{12}. \quad (78)$$

19 Constraints from unsuccessful mechanisms

The following negative results constrain where a reduced anomaly could come from.

- Fixed Gaussian localization produces Gaussian overlaps and does not select σ_0 or $M^{-1/6}$.

- Canonical Gram orthogonalization reconstructs an ordinary hierarchy derivative and does not produce $M^{-1/6}$.
- Ordinary harmonic Dirichlet-to-Neumann response does not produce $A^{1/12}$.
- A stationary scalar Jacobi recursion produces square-root band-edge structure and does not select the exponent $1/3$.
- Self-adjoint running Gaussian geometry does not produce a net hierarchy orientation.
- Separable spatial Clifford multiplicity cancels from α_H and does not yield $\nu_H = 1/12$.

A reduced coefficient, if it exists, must come from additional transition-active innovation structure, nonuniform K^{-2} weighting, a nonseparable projector, boundary or domain data not represented by the unilateral benchmark, or other structure in the Master Operator. None of these is assumed here.

20 Structural-anomaly proposition

Proposition 1 (Structural hierarchy anomaly and projected scaling) *Consider a semi-infinite nested hierarchy with orientation operator Γ_h , transition operator V , innovation operator K , and finite-grade projector Π_N . Suppose α_H in (50) exists, and suppose $N(\rho) \sim \rho^D$. Assume the positive innovation-frame transport is controlled by the square root convention (52). Then the associated frame exponent is $\nu_H = (D/2)|\alpha_H|$.*

If the innovation boundary problem is represented by the weighted half-line operator with branches $\beta_{\pm} = 1 \pm 2\nu_H$, its flat-measure form has Bessel index $\nu_B = \nu_H$. If, in addition, the transition-accessible innovation spectrum is effectively one-dimensional and the inverse boundary response supplies the innovation coupling, then $J(M) \sim M^{-4\nu_H}$. Within $0 < 4\nu_H < 1$, if the nonanalytic Feshbach term dominates, $\Sigma(E, p) \sim p^2 E^{-4\nu_H}$ and

$$E \sim p^{2/(1+4\nu_H)} = p^{2/(1+2D|\alpha_H|)}. \quad (79)$$

Remark 1 *The proposition is conditional on the stated hierarchy asymptotics, transport convention, boundary realization, innovation spectral density, p^2 factor, and dominance of the non-analytic Feshbach term. It is not a claim that every HSMT realization satisfies these assumptions.*

21 The open coefficient problem

The fractional-scaling problem reduces to one structural quantity. For the candidate chain $\nu_B = 1/12$, the anomaly must satisfy $|\alpha_H| = 1/(6D)$. The minimal semi-infinite hierarchy instead gives $|\alpha_H^{\min}| = 1/2$. The candidate value cannot be attributed solely to the existence of a lowest grade.

The outstanding problem is to derive α_H from the full, potentially nonseparable HSMT Master Operator. Possible sources of a reduced anomaly include multiple innovation families with partially cancelling boundary responses, nonuniform innovation gaps, a projector that does not factor between hierarchy and spatial sectors, boundary or domain data absent from the unilateral benchmark, and additional operator structure. These possibilities are not assumed.

22 Discussion

Geometric containment, state overlap, and dynamical coupling are distinct. Containment is represented by ι_N or J_N . Overlap is represented by $S_{NM} = \langle N|M \rangle$. Dynamical communication is represented by V_N . None of these objects should be identified automatically with the others.

The Gaussian kernel describes hierarchy localization and transition proximity. It does not determine the fractional interface exponent.

A semi-infinite hierarchy has a global asymmetry without non-Hermitian microscopic dynamics. The bulk transition operator may remain locally reversible while a lowest grade produces a finite boundary defect. In the minimal model,

$$\frac{\text{finite boundary defect}}{\text{linearly accumulating hierarchy content}} \sim \frac{1}{N}. \quad (80)$$

Accumulation gives $\sum_{n \leq N} n^{-1} \sim \ln N$, and the transport convention converts the logarithm into a power. This explains the form of an anomalous hierarchy exponent. It does not fix the value. The minimal model gives a substantially larger anomaly than the candidate Bessel index $1/12$.

None of these results requires interpreting ρ as an additional ordinary spatial coordinate. The fractional boundary operator, when assumed, describes response in innovation space. Its influence on physical dynamics appears only after projection through the Feshbach self-energy.

23 Conclusion

Nested embeddings $J_N : \mathcal{H}_N \rightarrow \mathcal{H}_{N+1}$ and the innovation decomposition $\mathcal{H}_{N+1} = J_N \mathcal{H}_N \oplus \mathcal{K}_{N+1}$ give a transition geometry for HSMT. The covariant difference $\nabla_N \psi = \psi_{N+1} - J_N \psi_N$ separates inherited change from new structure. The operator $A_N = V_N K_{N+1}^{-2} V_N^\dagger$ measures transition-active innovation response.

The radial Dirac structure supplies a conjugate orientation pair. Hermiticity does not select a preferred bulk direction. A semi-infinite hierarchy does possess a finite lower-boundary defect. In the minimal Pauli–Jacobi benchmark,

$$\mathfrak{p}_N = -\frac{1}{2N-1}, \quad \alpha_H^{\min} = -\frac{1}{2}. \quad (81)$$

The resulting N^{-1} imbalance can be converted, by an additional transport convention, into power-law frame dressing. Under further assumptions the conditional chain is

$$\alpha_H \longrightarrow \nu_H = \nu_B \longrightarrow \beta_\pm \longrightarrow c_{\text{IS}} \longrightarrow v(M) \longrightarrow J(M) \longrightarrow \Sigma(E, p) \longrightarrow z.$$

In particular, $\nu_B = 1/12$ would imply $c_{\text{IS}} = 35/144$, $J(M) \sim M^{-1/3}$, and $E \sim p^{3/2}$. This paper does not derive $\nu_B = 1/12$. It shows that these quantities form one conditional chain and isolates their selection in the calculation of α_H . The minimal hierarchy yields $|\alpha_H| = 1/2$, so the candidate value is not automatic.

The next mathematical problem is to determine the boundary-to-bulk innovation response of the full HSMT operator and to establish whether its structural hierarchy anomaly selects a nonzero fractional fixed value.

A Exact unequal-width Gaussian overlap

For normalized Gaussians of curvatures ω_N and ω_M , completing the square gives

$$\langle g_N | g_M \rangle = \left(\frac{2\sqrt{\omega_N \omega_M}}{\omega_N + \omega_M} \right)^{1/2} \exp \left[-\frac{\omega_N \omega_M}{2(\omega_N + \omega_M)} (\rho_N - \rho_M)^2 \right]. \quad (82)$$

If $\omega_{N+1} = \omega_N(1 + \varepsilon_N)$ and $x_N = \omega_N a_N^2$, then

$$\ln S_N = -\frac{x_N}{4} - \frac{x_N}{8} \varepsilon_N + \frac{x_N - 1}{16} \varepsilon_N^2 + O(\varepsilon_N^3).$$

For $\omega_N \sim N^s$, one has $\varepsilon_N = s/N + O(N^{-2})$. Running widths generate $O(N^{-1})$ corrections. Those corrections are orientation-even unless additional structure distinguishes the conjugate transition directions.

B Unequal-width derivative matrix element

The neighboring derivative matrix element is

$$\langle g_N | \partial_\rho | g_{N+1} \rangle = \frac{\omega_N \omega_{N+1}}{\omega_N + \omega_{N+1}} a_N S_N. \quad (83)$$

For $\omega_{N+1} = \omega_N(1 + \varepsilon_N)$,

$$B_N = \frac{\omega_N a_N}{2} e^{-x_N/4} \left[1 + \frac{4 - x_N}{8} \varepsilon_N + O(\varepsilon_N^2) \right].$$

Thus $x_N = 4$ is a stationary point of this first-order correction. No physical selection of $x_N = 4$ is implied.

C Pairwise innovation coupling

For equal-width neighboring packets, the pairwise orthogonal innovation state is

$$|k_{N+1}\rangle = \frac{|g_{N+1}\rangle - S|g_N\rangle}{\sqrt{1 - S^2}}.$$

The corresponding matrix element is $v_N^{\text{innov}} = (D_{N,N+1} - S D_{NN}) / \sqrt{1 - S^2}$. For the harmonic radial benchmark,

$$D_{N,N+1} - S D_{NN} = \frac{\omega a S}{2} (\sigma^2 - i\sigma^1). \quad (84)$$

The common m_0 term and the common background value of W cancel. The spin-averaged squared coupling is

$$|v^{\text{innov}}|_{\text{av}}^2 = \frac{\omega^2 a^2 S^2}{2(1 - S^2)}. \quad (85)$$

With $x = \omega a^2$, $S = e^{-x/4}$, and benchmark gap $\Delta^2 \simeq 2\omega$,

$$\chi_{\text{GS}}(x) = \frac{x}{4(e^{x/2} - 1)}. \quad (86)$$

This pairwise residual is not the canonical innovation of the complete semi-infinite hierarchy. The full inherited subspace must be removed when all preceding grades are retained.

D Full-history Gaussian innovation

For a one-sided sequence of Gaussian packets, let G_N be the Gram matrix of the first N packets. The squared norm of the sequential Gram-Schmidt residual is $E_N = \det G_{N+1} / \det G_N$. For the Toeplitz symbol g_x , the strong Szegő limit gives

$$E(x) = \exp \left[\frac{1}{2\pi} \int_0^{2\pi} \ln g_x(k) dk \right] = \prod_{n=1}^{\infty} (1 - e^{-nx/2}). \quad (87)$$

For $x \rightarrow 0$,

$$E(x) \sim \sqrt{\frac{4\pi}{x}} \exp \left[-\frac{\pi^2}{3x} + \frac{x}{48} \right].$$

In the strongly overlapping regime, the genuinely new orthogonal component becomes exponentially small after projection against the complete inherited history. This is distinct from the pairwise residual $1 - S^2 \sim x/2$.

E Weighted-branch intertwiner

On $\mathcal{H}_\beta = L^2((0, \infty), \rho^\beta d\rho)$, the flattening map $U_\beta f = \rho^{\beta/2} f$ converts L_β to $-\partial_\rho^2 - \beta(2-\beta)/(4\rho^2)$. The branches β and $2-\beta$ share this flat normal form. The intertwiner is $T_{\beta \rightarrow 2-\beta} = U_{2-\beta}^{-1} U_\beta$, so

$$(Tf)(\rho) = \rho^{\beta-1} f(\rho). \quad (88)$$

For $\beta_\pm = 1 \pm 2\nu$, the reciprocal branch dressing is $\rho^{\pm 2\nu}$. At $\nu = 1/12$ this is $\rho^{\pm 1/6}$. The reciprocal distortion follows once the Bessel index is specified. It does not select $\nu = 1/12$.

F Endpoint behavior and domain dependence

For $H_\nu = -\partial_\rho^2 + (\nu^2 - 1/4)\rho^{-2}$, the Frobenius solutions behave as $\psi_\pm(\rho) \sim \rho^{1/2 \pm \nu}$. For $0 < \nu < 1$, both are locally square integrable near $\rho = 0$, so the endpoint is limit-circle and a self-adjoint boundary condition is required. A generic boundary relation $B = \Lambda^{2\nu} A$ introduces a scale Λ . The scale-invariant limits correspond schematically to $A = 0$ or $B = 0$. These orientations are compatible with the conjugate weighted branches. Scale invariance does not determine ν .

G Exponent audit

The structural hierarchy anomaly is $\alpha_H = \lim_{N \rightarrow \infty} N \mathfrak{p}_N$, when the limit exists. If $N(\rho) \sim \rho^D$ and the transport convention (52) is adopted, then $\nu_H = (D/2)|\alpha_H|$. Set $h = 2\nu_H = D|\alpha_H|$. The weighted branches, if imposed, are $\beta_\pm = 1 \pm h$. Then

$$c_{\text{IS}} = \frac{1 - h^2}{4} = \frac{1}{4} - \nu_H^2, \quad \nu_B = \sqrt{\frac{1}{4} - c_{\text{IS}}} = \nu_H.$$

The inverse boundary response is $v(M) \sim M^{-2\nu_B}$, so $|v(M)|^2 \sim M^{-4\nu_B}$. For a flat one-dimensional innovation density, $J(M) \sim M^{-4\nu_B}$ and $\Sigma(E, p) \sim p^2 E^{-4\nu_B}$. The dispersion exponent is

$$z = \frac{2}{1 + 4\nu_B} = \frac{2}{1 + 2D|\alpha_H|}.$$

For the conditional input $\nu_B = 1/12$, one obtains $h = 1/6$, $\beta_{\pm} = 5/6, 7/6$, $c_{\text{IS}} = 35/144$, $v(M) \sim M^{-1/6}$, $J(M) \sim M^{-1/3}$, $\Sigma \sim p^2 E^{-1/3}$, and $z = 3/2$.

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